Jin Tao

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Education

Technische Universität Berlin — Berlin, Germany
M.Sc. in Computational Engineering Science; GPA: 1.3 (1.0 best) — *Feb. 2025 – Present*

Technische Universität Berlin — Berlin, Germany
M.Sc. in Vehicle Engineering — *Sep. 2020 – Feb. 2025*

Hainan University — Hainan, China
B.Sc. in Vehicle Engineering — *Sep. 2015 – June 2019*

Experience

Researcher (previously Working Student) — May 2026 – Present
Center for Tangible Artificial Intelligence and Digitalization (ZEKI) — *Berlin, Germany*

- Develop a full-stack robotics platform on TurtleBot integrating SLAM, navigation, semantic mapping, autonomous exploration, and vision-language models, with an AI agent that interprets user requests into robot actions.
- Develop a multi-robot collaboration system for disaster response, integrating UGVs and UAVs for collaborative mapping and task coordination through a central device.

Research Assistant — Oct. 2025 – Apr. 2026
Robotics and Biology Laboratory (RBO) — *Berlin, Germany*

- Designed and built an acoustic gripper from scratch, using acoustic signatures to identify grasped objects.
- Improved the software pipeline for a Franka manipulator by resolving a persistent impedance-controller bug that caused abnormal oscillations at higher impedance settings.

Working Student — Sep. 2024 – Oct. 2025
Mbition GmbH — *Berlin, Germany*

- Developed a full-stack software system for monitoring test vehicles and managing vehicle bookings.

Software Engineer — Oct. 2022 – Mar. 2024
TU Berlin Formula Student Team - FaSTTUBe — *Berlin, Germany*

- Implemented model predictive control (MPC) in C++ using the FORCESPRO solver for the team's race car, with a minimum-lap-time objective and constraints on track boundaries and vehicle dynamics.
- Developed ROS 2 trajectory tracking using pure pursuit and MPC, and tested the system in the lightweight racing simulator Live for Speed.
- Developed a steering control system using cascaded current, speed, and position PID loops to achieve fast, stable tracking of commanded steering positions.

Team Leader — May 2018 – Nov. 2019
Hainan University Formula Student Team — *Hainan, China*

- Designed and built the vehicle's double-wishbone suspension, applying vehicle dynamics principles to determine its geometry.
- Led team organization and coordination, and competed as the endurance-event driver at Formula Student China 2019.

Achievements and Projects

TaoRobot: Autonomous Driving Full Stack — June 2026 – Present

- Open-Source Personal Project | github.com/JinTTTT/taorobot; demo available in the GitHub repository.
- Independently developing a lightweight autonomous-driving stack from scratch in ROS 2, with custom SLAM, localization, planning, and control algorithms implemented as modular nodes.
- Implemented pose-graph SLAM, particle-filter localization, A* path planning with path smoothing, and pure pursuit trajectory tracking; integrated the stack in Gazebo simulation.

Magnetic Joint-State Estimation — Mar. 2026 – Sep. 2026

- Proposed and developed a ball-joint orientation estimation system using two Adafruit magnetic sensors costing less than €7 each and a magnet to estimate yaw, pitch, and roll.
- Evaluated orientation accuracy against an Xsens MTi-630 IMU, achieving comparable accuracy under the tested conditions without long-term drift.
- Preparing a conference paper with the project team.

Machine Learning for Visuo-Tactile Sensor Fusion Mar. 2026 – Sep. 2026

- Developed training and evaluation infrastructure for estimating in-hand object pose from RGB images and tactile measurements, using ground-truth pose annotations.
- Implemented and compared ten existing methods, then developed a transformer-based fusion model that matched the best evaluated baseline's pose-estimation accuracy.

Master's Thesis: Autonomous Shuttle Pedestrian Following Aug. 2024 – Jan. 2025

- Developed and deployed a pedestrian-following system on a real autonomous shuttle, using A* path planning for trajectory generation, high-level pure pursuit control for trajectory tracking, and low-level cascaded PID control.
- Integrated YOLO-based gesture detection with commands to stop, resume driving, and adjust following distance.

GPU-Accelerated Visual SLAM and Navigation Integration Mar. 2025 – Sep. 2025

- Deployed GPU-accelerated visual SLAM on a Jetson Xavier-based mobile robot and integrated its outputs with Nav2 by projecting 3D map points into a 2D occupancy grid.
- Reduced CPU utilization by 40% while increasing GPU utilization by 20%, and improved navigation through faster map–odometry updates and more accurate localization.

Skills

Languages: English (advanced), German (intermediate), Chinese (native)
Programming: Python, C/C++, MATLAB/Simulink
Planning and Control: Model predictive control (MPC), FORCESPRO, A* path planning, pure pursuit, cascaded PID control
Simulation: Live for Speed
Hardware: Arduino, ESP32, Raspberry Pi, NVIDIA Jetson Nano/Xavier; soldering, PCB design and debugging
Frameworks: ROS, ROS 2, Nav2, TensorFlow, PyTorch, TensorRT, YOLO, LangChain, LangGraph, Ollama, llama.cpp
Development Tools: Git, Docker

Selected Coursework (Grades 1.0–1.3; 1.0 Best)

Robotics: 1.0 | **Applications of Robotics and Autonomous Systems**: 1.0
Sensor Systems: 1.3 | **Machine Learning 1**: 1.3 | **Motion Planning**: 1.3
Numerical Analysis I for the Engineering Sciences: 1.0